

All My Work

Your saved learning record for Genetic Diversity in Populations.

1 completed activity run. Required checks and formative practice are recorded separately.

Unfinished work: 0 required check(s), 1 practice run(s). Resume them from their original sections; they are not counted as completed.

Writing and flash-card self-assessments are not automatically graded. Completion is not a mastery certificate.

Mixed practice

9/17/2026, 8:43:59 PM • Elapsed 0m 1s • Formative practice

Review submitted responses and attempts

First-try result: 7/7. Later corrections are recorded separately.

Which term fits this definition? A model in which allele frequencies remain constant and genotype frequencies follow p^2 , $2pq$ and q^2 under stated conditions.

Self-report: Study again.

Reference: Hardy–Weinberg equilibrium

Name the term described: The introduction of new sequence variants through mutation.

Attempt 1: mutation input — Correct.

Reference: mutation input

Which term matches this description? A change in allele frequencies in a population across generations.

Options shown: genetic variation · microevolution · population · gene pool

Attempt 1: microevolution — Correct.

Reference: microevolution

What does letter C identify?

Options shown: Heterozygote frequency · Expected carrier count · Recessive allele frequency · Homozygous dominant frequency · Other allele frequency · Observed aa fraction

Attempt 1: Other allele frequency — Correct.

Reference: Other allele frequency

Select the TWO accurate statements about populations, gene pools and genetic variation.

Options shown: A gene pool is the set of organs inside one organism. · Not all phenotypic variation is genetic. · Any increase in population size is automatically an allele-frequency change. · A population boundary must be stated before counting frequencies.

Attempt 1: true-2 → true-1 — Correct.

Reference: Not all phenotypic variation is genetic. · A population boundary must be stated before counting frequencies.

Put these steps in order: Calculating expected genotype frequencies

Options shown: Check totals and interpret the result · Calculate expected genotype frequencies · Identify the observed or supplied quantity · Convert frequencies to counts if requested · State the equilibrium and phenotype assumptions · Determine p and q

Attempt 1: Identify the observed or supplied quantity → State the equilibrium and phenotype assumptions → Determine p and q → Calculate expected genotype frequencies → Convert frequencies to counts if requested → Check totals and interpret the result — Correct.

Reference: Identify the observed or supplied quantity → State the equilibrium and phenotype assumptions → Determine p and q → Calculate expected genotype frequencies → Convert frequencies to counts if requested → Check totals and interpret the result

Assume equilibrium with $q = 0.20$ and $N = 250$. How many aa individuals are expected?

Self-report: Study again.

Reference: 10. $q^2 = 0.04$; $0.04 \times 250 = 10$.

Identify the term illustrated: Each Aa individual contributes one A and one a to the allele count.

Attempt 1: heterozygote contribution — Correct.

Reference: heterozygote contribution

A sample contains 10 AA, 20 Aa and 20 aa individuals. What is the a frequency?

Options shown: 0.40 · 0.80 · 0.60 · 0.20

Attempt 1: 0.60 — Correct.

Reference: 0.60

Explain genetic drift.

Self-report: Study again.

Reference: Random change in allele frequencies caused by chance sampling in finite populations.

Frayer work

population — Draft

Meaning: An edited explanation.

Key features: A specific learner explanation 2.

Example: A specific learner explanation 3.

Non-example / common confusion: A specific learner explanation 4.

gene pool — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

genetic variation — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

microevolution — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

genotype frequency — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

allele frequency — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

heterozygote contribution — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

frequency sum — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

Collected optional responses

Biotechnology and gene pools

The comparison alone cannot isolate the proposed chemical effect. Hormone production and receptor response are distinct steps.

Textbook question work

Draft and saved question responses. Optional and ungraded.

No textbook question work yet.

Supported and independent practice